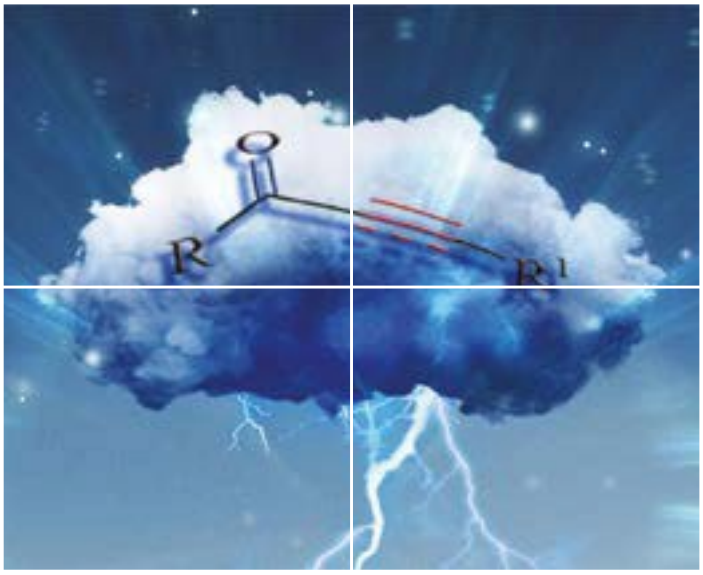
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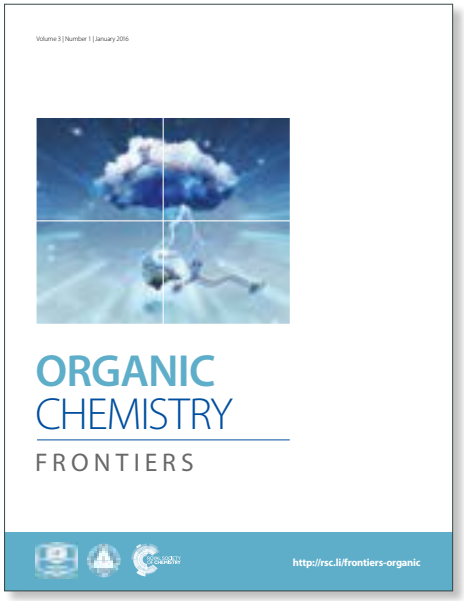
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Research Article

Efficient Trifluoromethylation via the Cyclopropanation of Allenes and Subsequent C-C Bond Cleavage

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As we know incorporation of trifluoromethyl group into organic molecules may significantly alter their physical and biological properties due to its high electronegativity, lipophilicity, excellent metabolic stability of the trifluoromethyl substituent. Thus, efficient method for introduction of trifuoromethyl group are of high current interest. On the other hand, vinylic cyclopropanes are a class strained compounds capable of undergoing ring-opening reaction with other molecules. Here CF_3-substituted vinylic cyclopropanes have been highly selectively formed by a copper-catalyzed cyclic trifluoromethylation of (4,4-disubstituted-2,3-butadienyl) malonates with Togni's reagent II, in which the trifluoromethyl group was installed to the middle carbon of the allene unit by applying 1,10-phenanthroline as the ligand. Such unique cyclopropanes successfully bring the trifluoromethyl group to other useful organic skeletons by selective cleavage of C–C bonds with an exclusive diastereoselectivity. Based on mechanistic studies, an allene radical addition, oxidation, and allylic substitution pathway has been proposed.

Trifluoromethylated compounds have been widely used in all aspects of chemistry, such as materials, pharmaceuticals, agrochemicals, and fine chemicals due to the high electronegativity, lipophilicity, and excellent metabolic stability of the trifluoromethyl substituent.[1-3] Of particular interest, compounds containing 2,2,2-trifluoroethylcyclopropane unit have been identified as selective androgen receptor modulators,[4a] *anti*-inflammatory agents,[4b,4c] immuno-modulators,[4b] and *anti*-tumor agents[4c] (Scheme 1). On the other hand, vinylcyclopropanes are the core structure of various pyrethroids such as pyrethrin,[4d,4e] permethrin,[4f,4g] cyhalothrin, [4h] and bifenthrin.[4i] Thus, we envisioned the structure of trifluoromethyl-substituted vinylcyclopropanes **2** and have been interested in developing methodologies for the efficient synthesis of this type of compound (Scheme 1). In addition, due to the high reactivity of the three-membered ring, such cyclopropanes **2** may also bring the trifluoromethyl group to other useful organic skeletons via the C-C bond cleavage reactions.[5] To the best our knowledge, there is no method for the construction of such CF_3-substituted vinylic cyclopropanes and no report on their related reactivity study. We proposed a trifluoromethylative cyclization of allenes containing a malonate unit for the efficient synthesis of **2**-type of compounds (Scheme 2D).[7,8] The challenge here would be the regioselectivity affording either the non-favored highly strained 3-membered products **2** or the most favored 5-membered products **3** as observed in the Pd-catalyzed cyclization of allenylmalonates with organic halides (Scheme 2C).[9] Herein, we report our recent observation on highly regioselective copper-catalyzed trifluoromethylation of (2,3-butadienyl)malonates using a hypervalent trifluoromethyl iodonium reagent, which allows for the exclusive formation of strained trifluoromethylated vinyl cyclopropanes **2**.

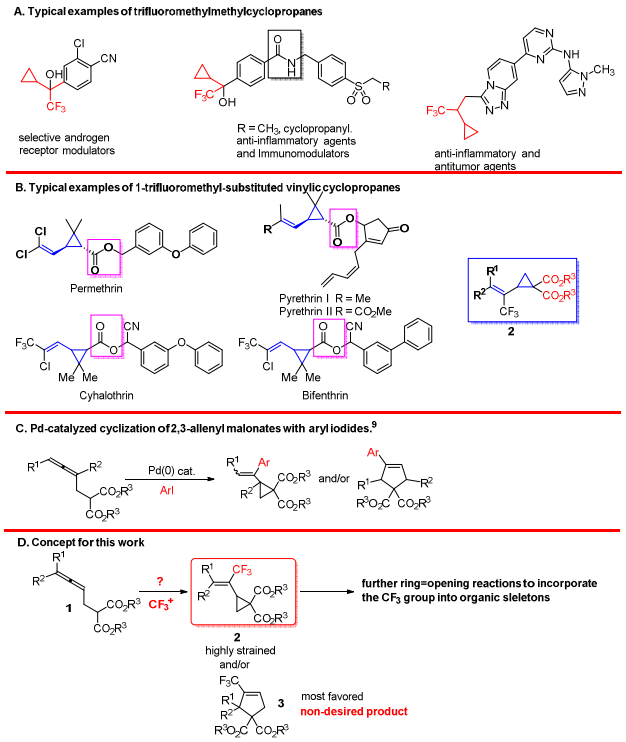
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Electronic supplementary information (ESI) available: Preparation and characterisation data as well as ^1H and ^{13}C NMR spectra of all compounds. See DOI: 10.1039/x0xx00000x

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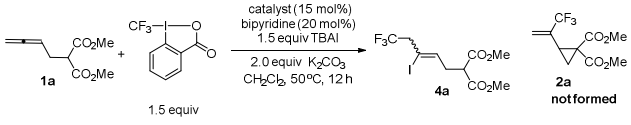
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Scheme 1. A and **B**. Some bioactive cyclopropanes containing a trifluoromethyl group; **C**. Pd-catalyzed cyclization of allenyl malonates with organic halides; **D**. Concept of introducing trifluoromethyl group via transition metal-catalyzed cyclopropanation

Our initial investigation started with the reaction of dimethyl 2-(buta-2,3-dienyl)malonate **1a** with Togni's reagent II in the presence of 5 mol% of $PdCl_2$ and 2 equiv of K_2CO_3 in DCM at 50 °C, however, no trifluoromethylation product **2a** was observed (Table 1, entry 1). Instead, highly regioselective iodo-trifluoromethylation product **3a** was detected albeit as a pair of *Z/E* stereoisomers.[10] This clearly indicated that in this simplest case the trifluoromethyl group was directed to the terminal position of the allene unit in dimethyl 2-(buta-2,3-dien-1-yl)malonate **1a**, excluding the possibility of forming the expected cyclized product **2a**. Cu(I) or Cu(II) catalysts afforded every similar results, still affording low yields of **4a** while the formation of **2a** was not detected (Table 1, entries 2-4).

Table 1. Effect of catalysts on iodo-trifluoromethylation of allene **1a**. [a]

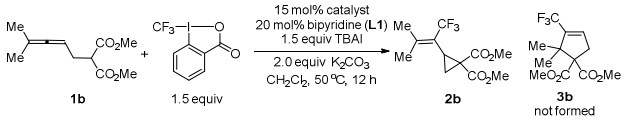


Entry	Catalyst	Yield of **4a** (%) (*Z*/*E*) [b]	Recovery of **1a** (%) [b]
1	$PdCl_2$ [c]	3 (2:1)	48
2	CuBr	13 (1.5:1)	26
3	$[Cu(CH_3CN)_4]PF_6$	20 (1.5:1)	16
4	$Cu(OAc)_2$ [d]	33 (2.3:1)	0

[a] Reaction conditions: Unless otherwise specified, the reaction was carried out using **1a** (0.2 mmol), Togni's reagent II (0.3 mmol), TBAI (0.3 mmol), K_2CO_3 (0.4 mmol), copper catalyst (0.03 mmol), and 2,2'-bipyridine (**L1**) (0.04 mmol) in 2 mL of CH_2Cl_2 under argon atmosphere. [b] Determined by ^{19}F NMR and ^{1}H NMR spectroscopy using $PhCF_3$ and CH_2Br_2 as the internal standards, respectively. [c] The reaction was carried out without bipyridine and TBAI. [d] The reaction was conducted on 1 mmol scale of **1a** with 2 equiv Togni's reagent II, K_3PO_4 (2 mmol) as the base, $Cu(OAc)_2$ (10 mol%), and 1,10-phenanthroline (**L2**) (0.2 mmol) as the ligand; reaction time was 22 h.

Thus, we envisioned to increase the steric hindrance of the terminal position of the allene unit in **1** for possible direction of the trifluoromethyl group to the allene middle carbon atom to form a π-allylic metal species, which would be followed by nucleophilic substitution to possibly afford the cyclized products **2** or **3**. When dimethyl 2-(buta-2,3-dien-1-yl)malonate **1a** was replaced with dimethyl 2-(4-methyl-2,3-pentadienyl)malonate **1b**, the reaction under the catalysis of $Cu(OAc)_2$, $CuCl_2$, $CuF_2{\cdot}H_2O$, $Cu(OTf)_2$ (Table 2, entries 1-4) did afford the designed trifluoromethylated vinylic cyclopropane **2b** exclusively in moderate yields and the formation of 5-membered ring **3b** was not observed. $Cu(OAc)_2$ gave the best results, affording **2b** in 65% yield with 28% recovery of **1b** (Table 2, entry 2). CuOAc could also catalyse the reaction with a slightly decreased yield (Table S1, entry 5). The reaction was even better with just 1.0 equiv of TBAI (Table 2, entry 6). Rather unexpectedly, reducing the catalyst loading of $Cu(OAc)_2$ from 15 mol% to 10 mol% improved the yield of **2b** from 65% to 69% (Table 2, entry 7). When we further reduced the loadings of catalyst and ligand, the yield dropped to 63% (Table 2, entry 8). By comparison, only 11% of product **2b** was obtained in the absence of copper complex (Table 2, entry 9).

Table 2. Effect of catalysts on trifluoromethylcylcopropanation of allene **1b.** [a]



Entry	Catalyst	Yield of **2b** (%) [b]	Recovery of **1b** (%) [b]
1	$Cu(OAc)_2$	65	28
2	$CuCl_2$	55	25
3	$CuF_2{\cdot}H_2O$	35	24
4	$Cu(OTf)_2$	47	53
5	CuOAc	42	0
6	$Cu(OAc)_2$ [c]	65	15
7	$Cu(OAc)_2$ [c,d]	69	16
8	$Cu(OAc)_2$ [c,e]	63	14
9	-	11	51

[a] Reaction conditions: Unless otherwise specified, the reaction was carried out using **1b** (0.2 mmol), Togni's reagent II (0.3 mmol), TBAI (0.3 mmol), K_2CO_3 (0.4 mmol), copper catalyst (0.03 mmol), and 2, 2'-bipyridine (**L1**) (0.04 mmol) in 2 mL of CH_2Cl_2 under argon atmosphere. [b] Determined by ^{19}F NMR and ^{1}H NMR spectroscopy

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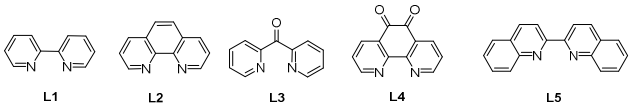
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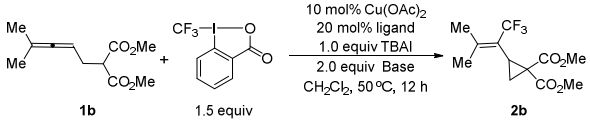
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using $PhCF_3$ and mesitylene or CH_2Br_2 as the internal standards. [c] The reaction was carried out using 1.0 equiv TBAI. [d] The reaction was carried out using 10 mol% $Cu(OAc)_2$. [e] The reaction was carried out with 5 mol% of $Cu(OAc)_2$, and 10 mol% of **L1.**



To further improve the yield, we applied the ligand effect by evaluating a series of bidentate ligands (Table 3): when 1,10-phenanthroline (**L2**) was used, the yield was further improved to 77% (Table 3, entry 2); the use of 1,10-phenanthroline-5,6-dione (**L4**) afforded the product in 42% yield; it is observed that the use of di-2-pyridyl ketone (**L3**) or 2,2'-biquinoline (**L5**) afforded **2b** in 12% and 27% yields, respectively (Table 3, entries 3 and 5). By contrast, in the absence of any ligand, the yield is much lower (Table 3, entry 6). Subsequently, we tested some bases with K_3PO_4 being the best (Table 3, entry 9). Further increasing the amount of K_3PO_4 led to a lower yield (Table 3, entry 10). To our delight, 2 equiv of Togni's II reagent led to full consumption of **1b**, affording **2b** in 87% yield (Table 3, entry 11). In the absence of a base, the reaction was poor (Table 3, entry 12). It should be noted that the yield dropped to 42% with full consumption of **1b** when the catalyst was replaced by CuOAc (Table 3, entry 13).

Table 3. Effect of ligands and bases on the reaction of trifluoromethylcylcopropanation.[a]

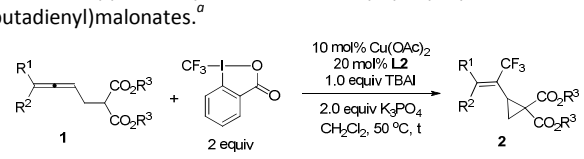


Entry	L	Base	Yield of **2b** (%)[b]	Recovery of **1b** (%)[b]
1	**L1**	K_2CO_3	69	16
2	**L2**	K_2CO_3	77	13
3	**L3**	K_2CO_3	12	27
4	**L4**	K_2CO_3	42	19
5	**L5**	K_2CO_3	27	26
6	-	**K_2CO_3**	10	34
7	**L2**	Na_2CO_3	63	17
8	**L2**	Cs_2CO_3	12	0
9	**L2**	K_3PO_4	82	9
10	**L2**	K_3PO_4[c]	76	12
11[d]	**L2**	**K_3PO_4**	**87**	**0**
12	**L2**	-	40	31
13[d,e]	**L2**	K_3PO_4	42	0

[a] Reaction conditions: Unless otherwise specified, the reaction was carried out using **1b** (0.2 mmol), Togni's reagent II (0.3 mmol), TBAI (0.2 mmol), base (0.4 mmol), $Cu(OAc)_2$ (0.02 mmol), and ligand (0.04 mmol) in 2 mL of CH_2Cl_2 under argon atmosphere. [b] Determined by ^{19}F NMR and ^1H NMR spectroscopy using $PhCF_3$ and mesitylene or CH_2Br_2 as the internal standards. [c] The reaction was carried out with 2.5 equiv of K_3PO_4. [d] The reaction was carried out with 2.0 equiv of Togni's reagent II. [e] The reaction was carried out with 10 mol% of CuOAc instead of $Cu(OAc)_2$.

With the optimized protocol in hand, we next turned to demonstrate the generality of this reaction. The results summarized in Table 4 show that this reaction indeed provides a straightforward entry to a series of trifluoromethylated vinyl cyclopropanes in moderate to good yields. Different substitutions at the terminal position of allene, such as methyl, ethyl, butyl, cyclobutyl, cyclopentyl and cyclohexyl could be compatible in this reaction, affording the corresponding trifluoromethylated vinyl cyclopropanes **2b–2j** in moderate to good yields (Table 4, entries 1-9); the substrate with sterically bulky isopropyl substituent also furnish the corresponding product **2k** in a moderate yield (Table 4, entry 10). Moreover, phenyl- and methyl-substituted malonate **1l** afforded corresponding product **2l** in 66% yield with a ratio of 1:3 for the corresponding stereoisomers (Table 4, entry 11). To further demonstrate the potential of this reaction, we carried out this reaction on gram scale under the standard conditions: when 1.0604 g of **1b** was used, 1.1061 g of **2b** was obtained in 79% yield.

Table 4. Copper-catalyzed trifluoromethylcyclopropanation of (2,3-butadienyl)malonates.[a]



Entry	R^1, R^2	R^3	t (h)	Yield of **2** (%)[b]
1	Me, Me	Me (**1b**)	12	86 (79[c])(**2b**)
2	-$(CH_2)_5$-	Me (**1c**)	24	84 (**2c**)
3	Me, Me	Et (**1d**)	13.5	82 (**2d**)
4	Me, Me	Bn (**1e**)	12	74 (**2e**)
5	Et, Et	Et (**1f**)	24	70 (**2f**)
6	*n*-Bu, *n*-Bu	Et (**1g**)	24	79 (**2g**)
7	-$(CH_2)_3$-	Et (**1h**)	17	64 (**2h**)
8	-$(CH_2)_4$-	Et (**1i**)	15	60 (**2i**)
9	-$(CH_2)_5$-	Et (**1j**)	13	74 (**2j**)
10[d]	*i*-Pr, *i*-Pr	Et (**1k**)	24	51 (**2k**)
11	Ph, Me	Me (**1l**)	12	66[e] (**2l**)

[a] Reaction conditions: Unless otherwise specified, the reaction was carried out using **1** (1 mmol), Togni's reagent II (2.0 mmol), TBAI (1 mmol), K_3PO_4 (2 mmol), $Cu(OAc)_2$ (0.10 mmol) and **L2** (0.20 mmol) in 7 mL of CH_2Cl_2 under argon atmosphere. [b] Isolated yield. [c] The reaction was carried out using **1b** (5 mmol, 1.0604 g), Togni's reagent II (10.0 mmol), TBAI (5 mmol), and $Cu(OAc)_2$ (0.5 mmol) and **L2** (1.0 mmol) in 35 mL of CH_2Cl_2 under argon atmosphere. [d] The reaction was carried out using 3 equiv of K_3PO_4 and 3 equiv of Togni's reagent II. [e] The reaction gave a pair of *Z/E* stereoisomers with a ratio of 1:3.

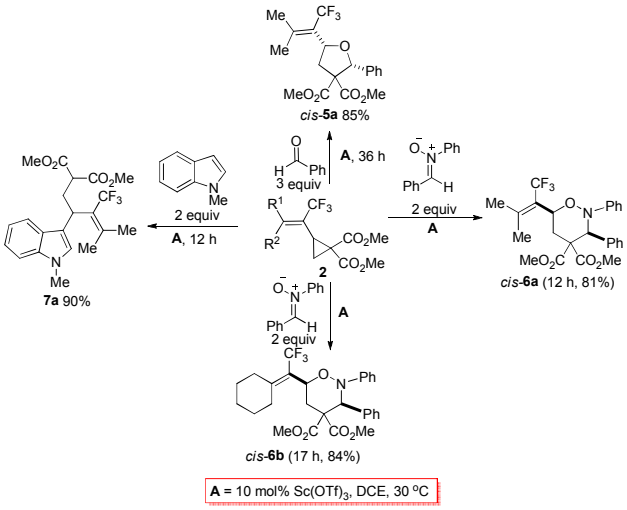
As stated in the introduction, one unique character of the strained three-membered ring is the selective cleavage of C–C bonds in cyclopropanes with easy incorporation of other molecules to afford a series of complex molecules bearing the trifluoromethyl group. After some screening of the reported Lewis acid catalysts for such transformation,[5,11] we observed that reactions catalyzed by 10

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mol% of $Sc(OTf)_3$ in DCE afforded the ring-opening products under very milder conditions: when trifluoromethylated vinylic cyclopropane **2b** was exposed to benzaldehyde, highly substituted tetrahydrofuran product *cis*-**5a**[12] was formed highly diastereoselectively in 85% yield; with nitrone, *cis* tetrahydro-1,2-oxazines *cis*-**6a**[12] and *cis*-**6b** were formed in 81% and 84% yields from **2b** and **2c**, respectively; the reaction of **2b** with *N*-methylindole afforded ring-opened functionalized indole product **7a**[12] in 90% yield (Scheme 2 and Fig. 1).



Scheme 2. $Sc(OTf)_3$-catalyzed reactions of **2**.

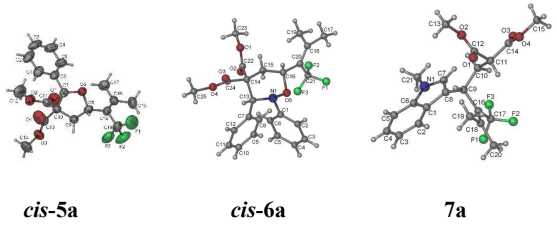
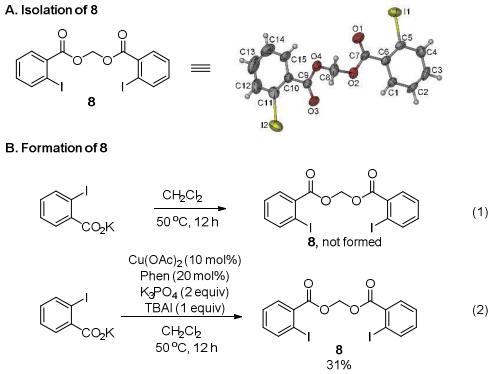
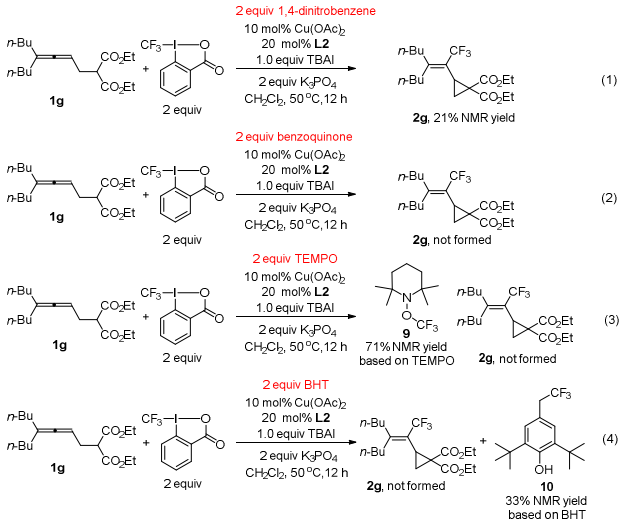


Fig. 1. ORTEP representation of *cis*-**5a,** *cis*-**6a**, and **7a**

During the study, we also identified the Togni's reagent II-based by-product by the X-ray diffraction study unambiguously as methylene bis(2-iodobenzoate) **8** (Scheme 3A).[12] Control experiments showed that in the absence of the copper complex, potassium 2-iodobenzoate didn't react with CH_2Cl_2 (Scheme 3B, eq. 1). When potassium 2-iodobenzoate was exposed under the standard conditions without the allene and K_3PO_4, compound **8** was formed in 31% yield. In order to further study the mechanism, radical scavengers were added under the standard reaction conditions (Scheme 4). With 1,4-dinitrobenzene, the reaction was somewhat suppressed to yield 21% of **2g** (Scheme 4, eq. 1). With benzoquinone, the trifluoromethylative cyclization reaction was completely shut down (Scheme 4, eq. 2). With TEMPO, the reaction didn't occur and the radical trapping TEMPO-CF_3 adduct **9** was formed in 71% yield (Scheme 4, eq. 3). With BHT, the trifluoromethylative product **2g** was not formed while a BHT-CF_3 adduct **10** was observed in 33% yield as judged by the analysis of the crude product comparing the signals with those reported in the literature[13] (Scheme 4, eq. 4). These results indicated that the reaction may proceed via a radical pathway at the beginning.



Scheme 3. Formation of by-product **8**: A. ORTEP representation. B. Control experiments.



Scheme 4. Radical trapping experiments.

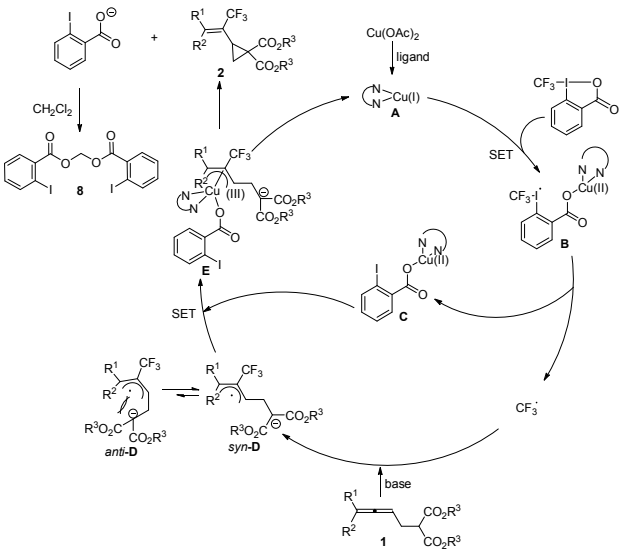
A mechanism was then proposed on the basis of the above results (Scheme 5). Initially, *in-situ* reduction[14] or disproportionation[15] of $Cu(OAc)_2$ forms the highly reactive Cu(I), which would coordinate with the ligand forming catalytically active copper(I) species **A**. Then a radical intermediate **B** could be generated by the reaction of **A** with Togni's reagent II, which would further release the CF_3 radical and (2-iodobenzoyloxy)copper (II) **C**. Allene 1 would be attacked by the trifluoromethyl radical and its nucleophilic unit would be deprotonated with the base to form the thermodynamically more stable π-allylic radical *syn*-**D**. The intermediate *syn*-**D** would further undergo oxidation with Cu (II) species **C** yielding π-allylic copper (III) intermediate E, which would undergo a intramolecular nucleophilic attack to release the cyclopropane products **2** and *o*-iodobenzoic acid anion associated with the regeneration of the catalytically active Cu (I) species **A**. The reaction of two molecules of *o*-iodobenzoic acid anion with CH_2Cl_2 would generate the isolated by-product **8**.[16] The unfavored formation of *anti*-**D** excludes the formation of 3-type of 5-membered ring. However, it should be noted that the mechanism requires more study and there may be other possibilities.

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Scheme 5. Proposed mechanism.

In conclusion, we have demonstrated an efficient copper-catalyzed efficient introduction of trifluoromethyl group into organic skeletons through the cyclization of allenes and C-C bond cleavage-based transformations via the formation of the strained trifluoromethylated vinyl cyclopropanes with an excellent regioselectivity under ambient conditions. Further studies in this area are ongoing in our laboratory.

Acknowledgement

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Notes and references

1 a) P. Kirsch, *Modern Fluoroorganic Chemistry* Wiley-VCH: Weinheim, **2004**. b) M. Shimizu, T. Hiyama, *Angew. Chem. Int. Ed.* **2005**, *44*, 214. c) K. Uneyama, *Organofluorine Chemistry* Blackwell: Oxford, U. K., **2006**. d) I. Ojima, *Fluorine in Medicinal Chemistry and Chemical Biology* Wiley-Blackwell: Chichester, U. K., **2009**. e) K. Müller, C. Faeh, F. Diederich, S*cience* **2007**, *317*, 1881. f) M. Hird, *Chem. Soc. Rev.* **2007**, *36*, 2070. g) K. L. Kirk, *Org. Process Res. Dev.* **2008**, *12*, 305. h) D. O'Hagan, *Chem. Soc. Rev.* **2008**, *37*, 308. i) R. Filler, R. Saha, *Future Med. Chem.* **2009**, *1*, 777. j) J. L. Aceña, Simón- A. Fuentes, S. Fustero, *Curr. Org. Chem*. **2010**, *14*, 928. k) C. Ni, J. Hu, *Chem. Soc. Rev.* **2016**, *45*, 5441.

2 For selected reviews, see: a) P. Jeschke, *ChemBioChem* **2004**, *5*, 570. b) S. Purser, P. R. Moore, S. Swallow, V. Gouverneur, *Chem. Soc. Rev.* **2008**, *37*, 320. c) D. Cahard, V. Bizet, *Chem. Soc. Rev.* **2014**, *43*, 135. d) E. P. Gillis, K. J. Eastman, M. D. Hill, D. J. Donnelly, N. A. Meanwell, *J. Med. Chem.* **2015**, *58*, 8315.

3 For selected recent reviews on trifluoromethylation, see: a) T. Furuya, A. S. Kamlet, T. Ritter, *Nature* **2011**, *473*, 470. b) O. A. Tomashenko, V. V. Grushin, *Chem. Rev.* **2011**, *111*, 4475. c) T. Liang, C. N. Neumann, T. Ritter, *Angew. Chem. Int. Ed.* **2013**, *52*, 8214. d) H. Egami, M. Sodeoka, *Angew. Chem. Int. Ed.* **2014**, *53*, 8294. e) E. Merino, C. Nevado, *Chem. Soc. Rev.* **2014**, *43*, 6598. f) J. Charpentier, N. Früh, A. Togni, *Chem. Rev.* **2015**, *115*, 650. g) C. Alonso, E. Marigorta de Marigorta, G. Rubiales, F. Palacios, *Chem. Rev.* **2015**, *115*, 1847. h) X. Pan, H. Xia, J. Wu, *Org. Chem. Front.* **2016**, *3*, 1163. For selected recent reports on trifluoromethylation of alkenes with CF_3^+, see: i) A. T. Parsons, T. D. Senecal, S. L. Buchwald, *Angew. Chem. Int. Ed.* **2012**, *51*, 2947. j) C. Feng, T.-P. Loh, *Angew. Chem. Int. Ed.* **2013**, *52*, 12414. k) Z. Feng, Q.-Q. Min, Zhao, H.-Y.; Gu, J.-W.; Zhang, X. *Angew. Chem. Int. Ed.* **2015**, *54*, 1270. l) W. Kong, N. Fuentes, A. García-Domínguez, E. Merino, C. Nevado, *Angew. Chem. Int. Ed.* **2015**, *54*, 2487. m) B. Sahoo, J.-L. Li, F. Glorius, *Angew. Chem. Int. Ed.* **2015**, *54*, 11577. n) M. Asano, R. Tomita, T. Koike, M. Akita, *J. Fluorine Chem.* **2015**, *179*, 83. o) Q. Lefebvre, N. Hoffmann, M. Rueping, *Chem. Commun.* **2016**, *52*, 2493. p) L. Wu, F. Wang, X. Wan, D. Wang, P. Chen, G. Liu, *J. Am. Chem. Soc.* **2017**, *139*, 2904. q) T. Kawamoto, R. Sasaki, A. Kamimura, *Angew. Chem. Int. Ed.* **2017**, *56*, 1342. For selected recent reports on trifluoromethylation of arenes with CF_3^+, see: r) K. Zhang, X.-H. Xu, F.-L. Qing, *J. Org. Chem.* **2015**, *80*, 7658. s) L. Li, X. Mu, W. Liu, Y. Wang, Z. Mi, C.-J. Li *J. Am. Chem. Soc.* **2016**, *138*, 5809. t) J. Lin, Z. Li, J. Kan, S. Huang, W. Su, Y. Li, *Nat. Commun.* **2017**, *8*, 14353. doi: 10.1038/ncomms14353. For selected recent reports on trifluoromethylation of other compounds with CF_3^+, see: u) vinyl azides: Y.-F. Wang, G. H. Lonca, S. Chiba, *Angew. Chem. Int. Ed.* **2014**, *53*, 1067. v) 3-alkenoic acids: Z. He, P. Tan, J. Hu, *Org. Lett.* **2016**, *18*, 72. w) enol triflates: X. Su, H. Huang, Y. Yuan, Y. Li, *Angew. Chem. Int. Ed.* **2017**, *56*, 1338.

4 (4) a) A. L. Handlon, L. T. Schaller, L. M. Leesnitzer, R. V. Merrihew, C. Poole, J. C. Ulrich, J. W. Wilson, R. Cadilla, P. Turnbull, *ACS Med. Chem. Lett.* **2016**, *7*, 83. b) B. J. M. G. Cals, S. B. Nabuurs, *PCT Int. Appl.* **2015**, WO 2015082533 A1 20150611. c) R. Kaul, E. J. McEachern, J. Sun, D. J. Vocadlo, Y. Zhou, Y. Zhu, *PCT Int. Appl.* **2015**, WO 2015095963 A1 20150702. d) F. B. Laforge, W. F. Barthel, *J. Org. Chem.* **1947**, *12*, 199. e) L. Crombie, M. Elliott, S. H. Harper, H. W. B. Reed, *Nature* **1948**, *162*, 222. f) A. H. Glickman, T. Shono, J. E. Casida, J. J. Lech, *J. Agric. Food Chem.* **1979**, *27*, 1038. g) G. W. Ivie, L. M. Hunt, *J. Agric. Food Chem.* **1980**, *28*, 1131. h) P. D. Bentley, R. Cheetham, R. K. Huff, R. Pascoe, J. D. Sayle, *Pestic. Sci.* **1980**, *11*, 156. i) E. L. Plummer, R. R. Stewart, *J. Agric. Food Chem.* **1984**, *32*, 1116.

5 For selected reviews on the chemistry of cyclopropanes, see: a) H.-U. Reissig, R. Zimmer, *Chem. Rev.* **2003**, *103*, 1151. c) M. Rubin, M. Rubina, V. Gevorgyan, *Chem. Rev.* **2007**, *107*, 3117. d) F. Brackmann, A. de Meijere, *Chem. Rev*. **2007**, *107*, 4538. e) C. A. Carson, M. A. Kerr, *Chem. Soc. Rev.* **2009**, *38*, 3051. f) B.-L. Lu, L. Dai, M. Shi, *Chem. Soc. Rev.* **2012**, *41*, 3318.

6 For reviews, see: a) R. Zimmer, C. U. Dinesh, E. Nandanan, F. A. Khan, *Chem. Rev.* **2000**, *100*, 3067. b) J. A. Marshall, *Chem. Rev.* **2000**, *100*, 3163. c) A. S. K. Hashmi, *Angew. Chem. Int. Ed.* **2000**, *39*, 3590. d) X. Lu, C. Zhang, Z. Xu, *Acc. Chem. Res.* **2001**, *34*, 535. e) R. W. Bates, V. Satchareon, *Chem. Soc. Rev.* **2002**, *31*, 12. f) H. U. Reissig, W. Schade, G. M. O. Amombo, R. Pulz, A. Hausherr, *Pure Appl. Chem.* **2002**, *74*, 175. g) L. K. Sydnes, *Chem. Rev.* **2003**, *103*, 1133. h) S. Ma, *Acc. Chem. Res.* **2003**, *36*, 701. i) L. Brandsma, N. A. Nedolya, *Synthesis* **2004**, 735. j) L. Wei, H. Xiong, R. Hsung, *Acc. Chem. Res.* **2003**, *36*, 773. k) A. Hoffmann-Röder, N. Krause, *Angew. Chem. Int. Ed.* **2004**, *43*, 1196. l) S. Ma, *Chem. Rev.* **2005**, *105*, 2829. m) S. Ma, *Aldrichim. Acta* **2007**, *40*, 91. n) S. Ma, *Acc. Chem. Res.* **2009**, *42*, 1679. o) N. Krause, C. Winter, *Chem. Rev.* **2011**, *111*, 1994. p) S. Yu, S. Ma, *Angew. Chem. Int. Ed.* **2012**, *51*, 3074. q) J. Ye, S. Ma, *Acc. Chem. Res.* **2014**, *47*, 989.

7 (7) a) K. Tsuchii, M. Imura, N. Kamada, T. Hirao, A. Ogawa, *J. Org. Chem.* **2004**, *69*, 6658. b) N. Zhu, F. Wang, P. Chen, J. Ye,

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G. Liu, *Org. Lett.* **2015**, *17*, 3580. c) Y. Wang, M. Jiang, J. Liu, *Adv. Synth. Catal.* **2014**, *356*, 2907. d) Z. He, P. Tan, J. Hu, *Org. Lett.* **2016**, *18*, 72. d) J. Jacquet, S. Blanchard, E. Derat, M. D.-E. Murr, L. Fensterbank, *Chem. Sci.* **2016**, *7*, 2030. e) R. Tomita, T. Koike, M. Akita, *Chem. Commun.* **2017**, *53*, 4681.

8 For formation of γ-lactones: Q. Yu, S. Ma, *Chem. Eur. J.* **2013**, *19*, 13304.

9 a) S. Ma, Z. Zhao, *Org. Lett.* **2000**, *2*, 2495. b) S. Ma, N. Jiao, S. Zhao, H. Hou, *J. Org. Chem.* **2002**, *67*, 2837. c) S. Ma, N. Jiao, Q. Yang, Z. Zheng, *J. Org. Chem.* **2004**, *69*, 6463.

10 For selected examples on iodo-trifluoromethylation using Togni's reagent, see: a) P. G.; Janson, I. Ghoneim, N. O. Ilchenko, K. J. Szabó, *Org. Lett.* **2012**, *14*, 2882. b) N. O. Ilchenko, P. G. Janson, K. J. Szabó, *J. Org. Chem.* **2013**, *78*, 11087. c) H. Egami, Y. Usui, S. Kawamura, S. Nagashima, M. Sodeoka, *Chem. Asian J.* **2015**, *10*, 2190.

11 a) P. D. Phhlhaus, J. S. Johnson, *J. Org. Chem.* **2005**, *70*, 1057. b) I. S. Young, M. A. Kerr, *Angew. Chem. Int. Ed.* **2003**, *42*, 3023. c) M. A. Kerr, R. G. Keddy, *Tetrahedron Lett.* **1999**, *40*, 5671.

12 For X-ray single crystal data for *cis*-**5a**, *cis*-**6a**, **7a**, and **8**, see Supporting Information.

13 H. Egami, T. Ide, Y. Kawatoa, Y. Hamashima, *Chem. Commun.* **2015**, *51*, 16675.

14 For reduction, see: a) N. Matsuda, K. Hirano, T. Satoh, M. Miura, *Org. Lett.* **2011**, *13*, 2860. b) G. Li, C. Jia, K. Sun, Y. Lv, F. Zhao, K. Zhou, H. Wu, *Org. Biomol. Chem.* **2015**, *13*, 3207. c) X.-F. Xia, S.-L. Zhu, J.-B. Liu, D. Wang, Y.-M. Liang, *J. Org. Chem*. **2016**, *81*, 12482.

15 For disproportionation, see: a) X. Ribas, D. A. Jackson, B. Donnadieu, J. Mahía, T. Parella, R. Xifra, B. Hedman, K. O. Hodgson, A. Llobet, T. D. P. Stack, *Angew. Chem. Int. Ed.* **2002**, *41*, 2991. b) L. M. Huffman, S. S. Stahl, *J. Am. Chem. Soc.* **2008**, *130*, 9196. c) A. Casitas, A. E. King, T. Parella, M. Costas, S. S. Stahl, X. Ribas, *Chem. Sci.* **2010**, *1*, 326.

16 Y. Zhang, J. Han, Y. Xu, Y. Wei, *J. Chem. Res.* **2012**, *36*, 303.

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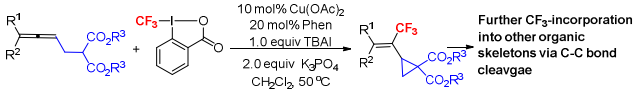
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